SUPERSTORE SALES & PROFIT DASHBOARD

Week 3 — Business Insights and Recommendations Report

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Programme: AnalystLab Africa — Data Analytics Internship
Project: Week 3 — Advanced Data Analysis, KPI Development & BI Dashboard
Tool: Microsoft Power BI

Date: August 2026

Executive Summary

This report extends the Week 2 analysis of the Superstore Sales & Profit Dashboard with deeper, business-focused investigation. Building on the Week 2 findings, this report examines monthly sales patterns, the relationship between discounting and profitability, and the specific drivers behind Furniture's weak profit margin. Three business problems were investigated in detail, supported by new dashboard visuals developed for Week 3. The findings reinforce and sharpen the Week 2 conclusions, and translate into a set of updated, evidence-based recommendations.

Business Problem Investigation

Problem 1 — Why does Furniture have high sales but low profit?

Investigation

The Furniture: Sales vs. Profit by Sub-Category chart shows that all four sub-categories (Chairs, Tables, Bookcases, Furnishings) generate meaningful sales, but their profit bars are minimal by comparison. Chairs has the highest sales among the four, yet its profit is only marginal. Tables is the most severe case, operating at a negative profit margin of approximately -9%, meaning the company loses money on every sale in that sub-category once costs are accounted for.

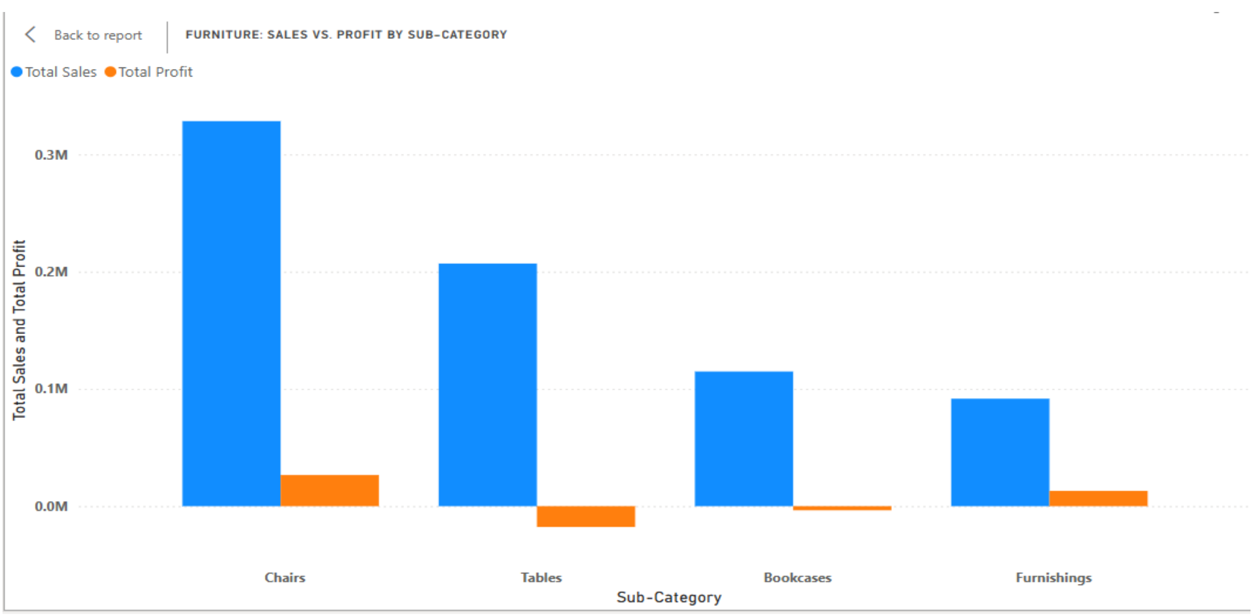
Root Cause

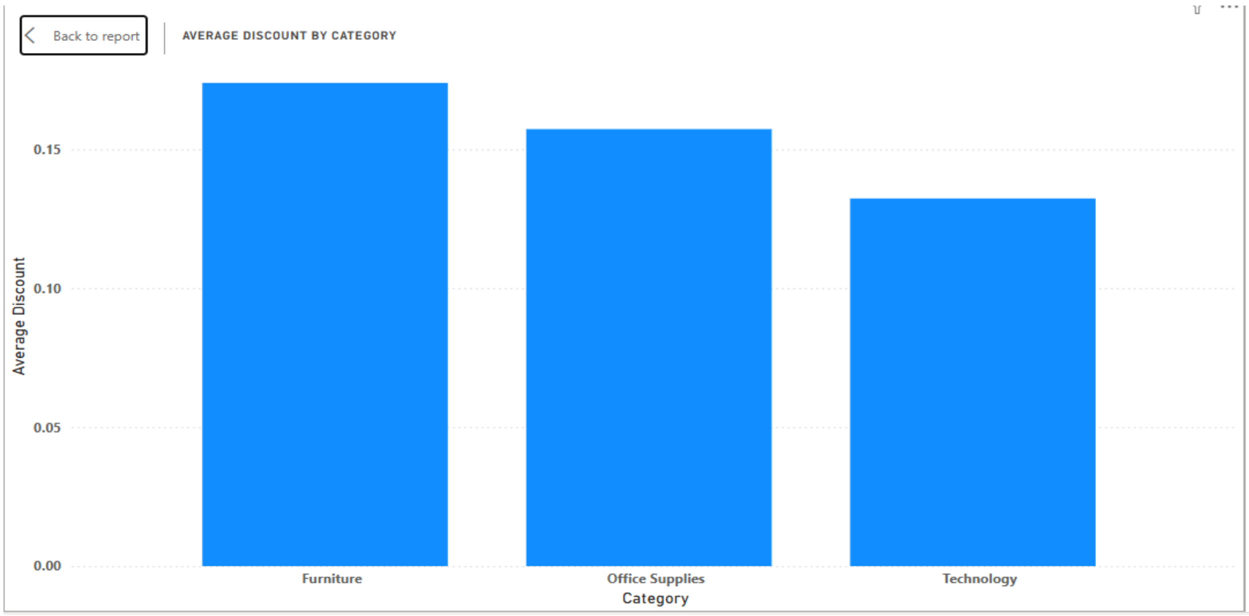
The Average Discount by Category chart shows Furniture carries a noticeably higher average discount than Office Supplies or Technology. Combined with the Discount vs. Profit by Sub-Category scatter chart — where several Furniture sub-categories cluster toward lower profit despite meaningful discount levels — this suggests discounting is a significant contributor to Furniture's weak profitability, not simply higher product costs.

Conclusion

Furniture is not a low-demand category: it sells well. The core issue is that aggressive discounting is eroding margins, particularly in Tables and Chairs.

Evidence from Dashboard





Problem 2 — Which months drive the strongest and weakest sales?

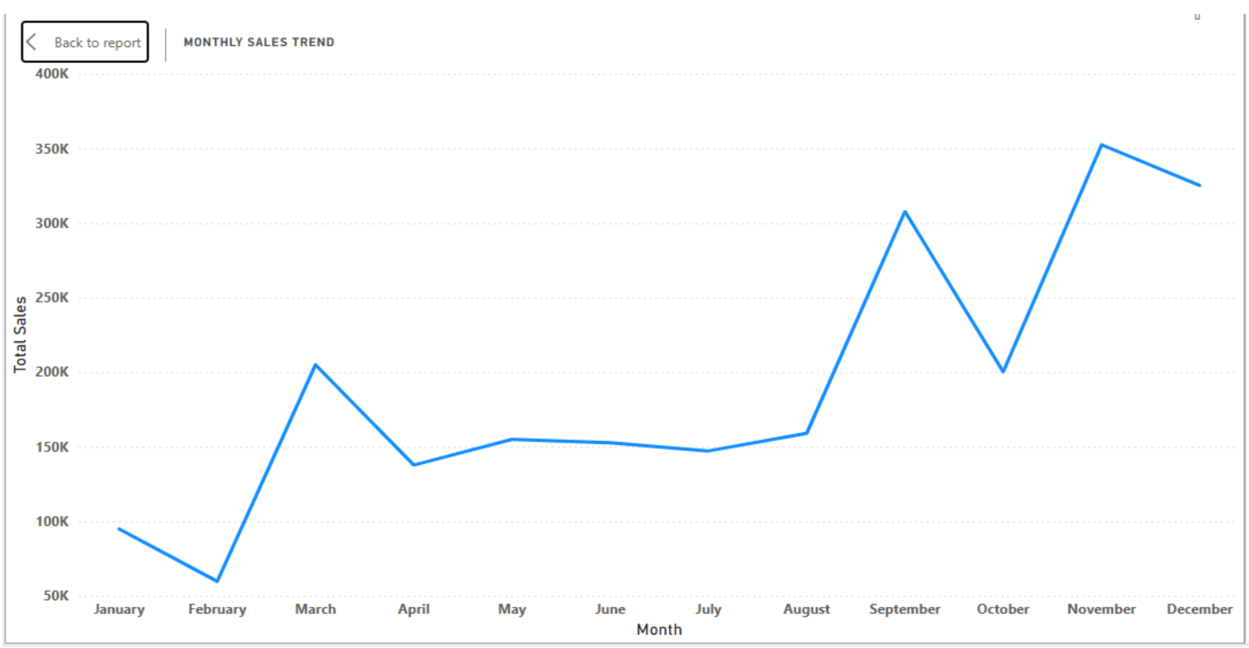
Investigation

The Monthly Sales Trend chart shows a clear pattern across the year. Sales start relatively low in January and February, remain fairly flat through the spring and summer months, then rise from September onward, peaking toward November and December.

Conclusion

The business experiences a clear end-of-year sales surge, most likely tied to the holiday shopping season, while January–February and the mid-year months are comparatively weaker. This has direct implications for inventory planning, marketing spend timing, and staffing, with resources best concentrated toward Q4 and targeted promotions considered for the slower early-year months.

Evidence from Dashboard



Problem 3 — How are discounts affecting profitability?

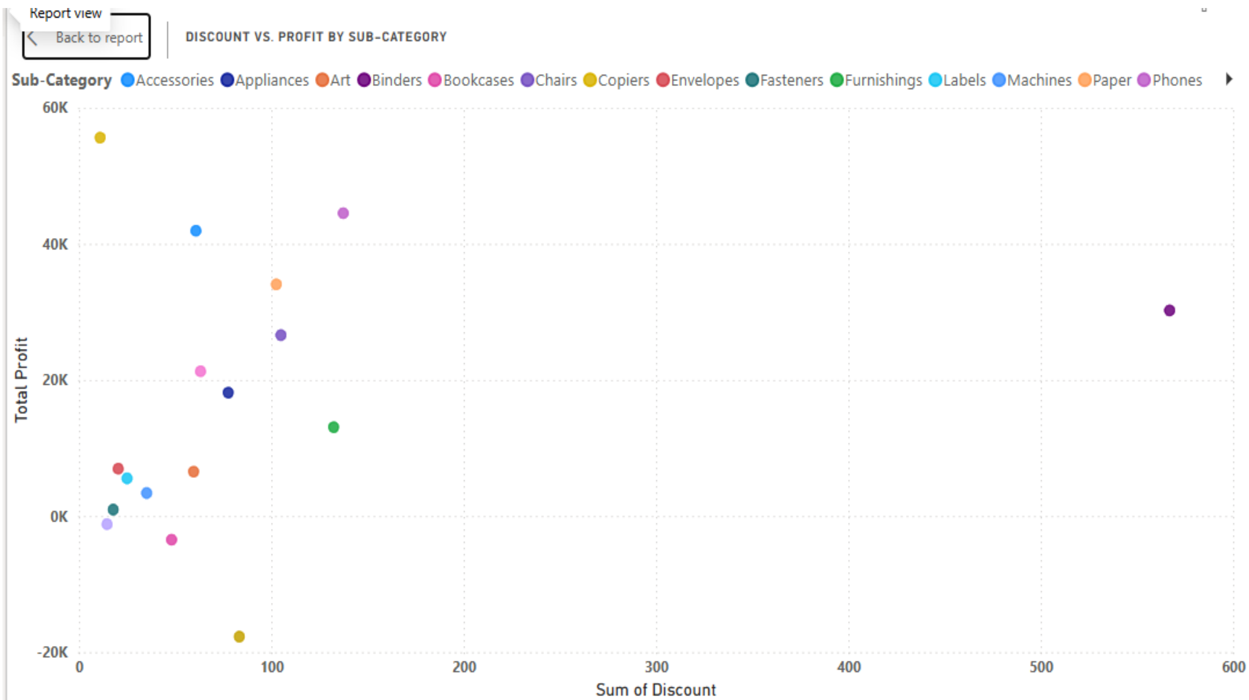
Investigation

The Discount vs. Profit by Sub-Category scatter chart shows that sub-categories with higher discount totals tend to sit toward lower profit, while sub-categories with minimal discounting show healthier profit outcomes. This pattern is reinforced by the Average Discount by Category chart, where Furniture — the category with the weakest profit margin identified in Week 2 — also carries the highest average discount of the three categories.

Conclusion

There is a visible relationship between discount level and reduced profitability. While discounting can help drive sales volume, the evidence suggests that beyond a certain point, discounts are cutting directly into profit rather than generating enough additional volume to compensate, particularly within Furniture.

Evidence from Dashboard



Updated Business Insights

Insight 6 — Specific products are driving the company's losses

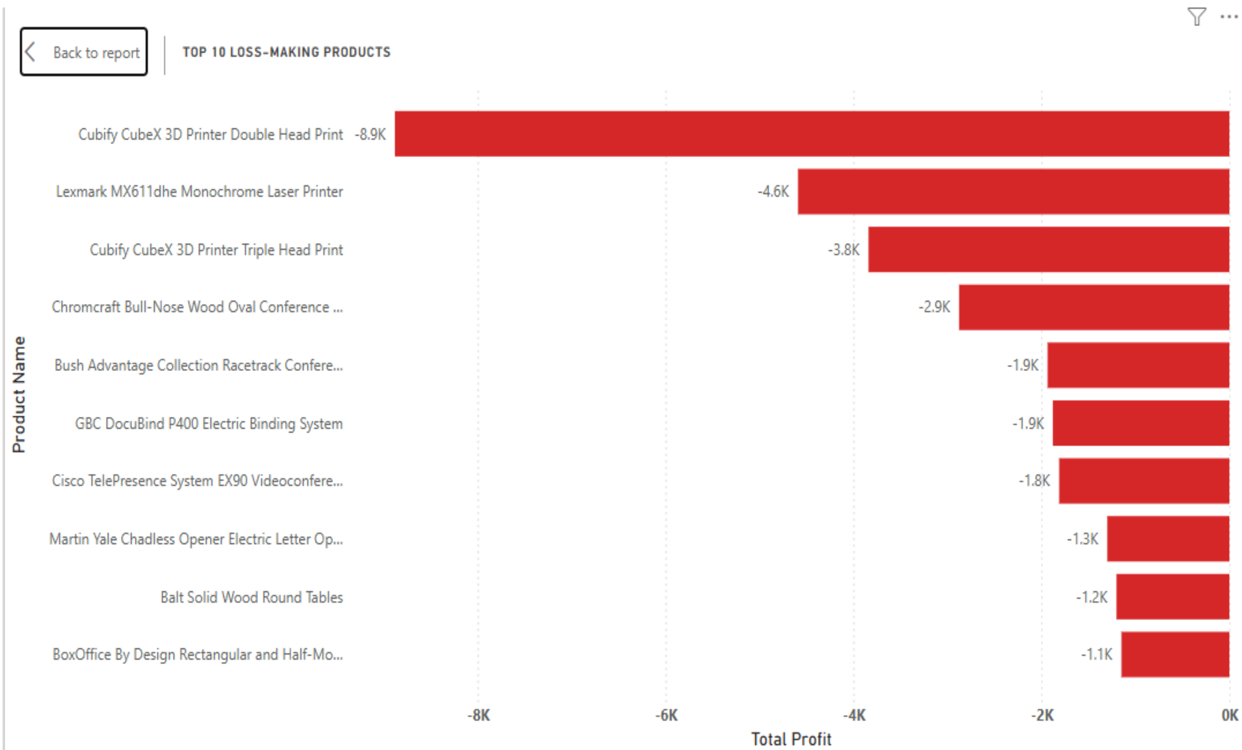
Finding

The Top 10 Loss-Making Products chart identifies specific individual products operating at a loss, led by a 3D printer product at approximately -$8.9K, followed by several other electronics and furniture items.

Business Interpretation

Losses are not evenly spread across the business, they are concentrated in a small number of identifiable products. This means the company can take targeted action (pricing review, discount limits, or discontinuation) on specific SKUs rather than broad category-wide changes.

Evidence from Dashboard



Insight 7 — Loss-making sub-categories exist beyond Furniture

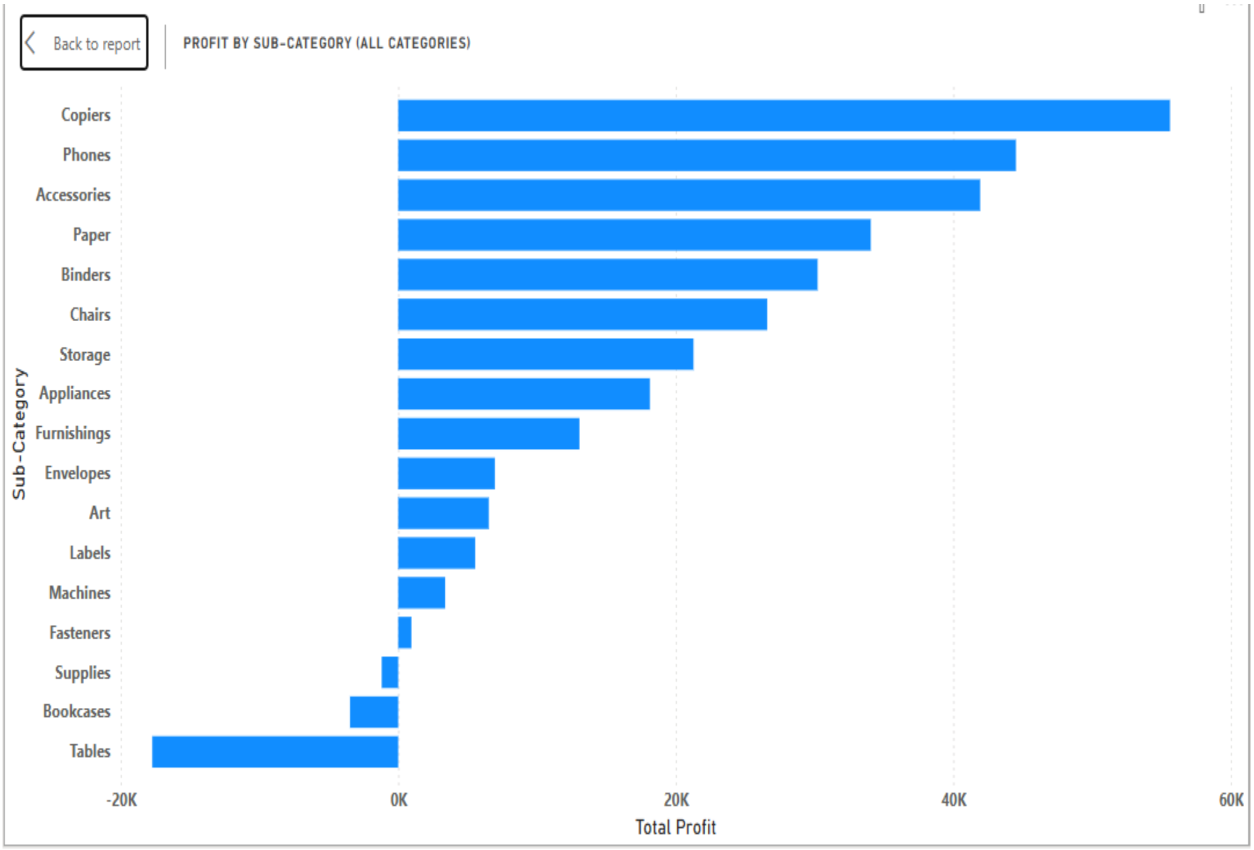
Finding

The Profit by Sub-Category (All Categories) chart shows that while Copiers, Phones, Accessories, Paper, and Binders remain solidly profitable, the overall sub-category ranking confirms that Furniture's Tables sub-category stands out as a persistent weak point company-wide, not just within its own category.

Business Interpretation

Viewed in the context of the whole business, Furniture's profitability issue is not a rounding error — it is a distinct, addressable problem worth direct management attention.

Evidence from Dashboard



Updated Recommendations

The following recommendations extend the Week 2 recommendations with findings from the Week 3 deeper analysis.

Recommendation 6 — Cap or restructure discounts on Furniture, especially Tables and Chairs

Management should review and likely reduce the maximum discount allowed on Furniture products, particularly Tables and Chairs, where high discount levels are most closely associated with low or negative profit.

Supporting Insight

Furniture carries the highest average discount of any category, and its sub-categories cluster toward lower profit on the Discount vs. Profit scatter chart.

Expected Impact

Tighter discount controls on Furniture could directly improve category profit margin without necessarily reducing sales volume significantly.

Recommendation 7 — Align inventory and marketing spend with seasonal demand

Management should increase inventory levels and marketing investment ahead of Q4 (September–December), while considering targeted promotions in the slower January–June period to smooth out demand.

Supporting Insight

The Monthly Sales Trend chart shows a consistent rise in sales from September through December, with earlier months comparatively weaker.

Expected Impact

Better alignment between resources and seasonal demand can reduce stockouts during peak months and improve sales during slower periods.

Recommendation 8 — Review the specific loss-making products directly

Management should conduct a targeted pricing and cost review of the specific products identified as the largest loss-makers, rather than addressing losses only at the category level.

Supporting Insight

The Top 10 Loss-Making Products chart identifies specific SKUs generating the largest losses, led by a 3D printer product at approximately -$8.9K.

Expected Impact

Directly addressing the worst-performing individual products offers a faster, more targeted path to improving overall profitability than category-wide changes alone.

Conclusion

The Week 3 analysis confirms and deepens the concerns first identified in Week 2. Furniture's profitability problem is closely tied to discounting practices, sales follow a clear seasonal pattern that should inform planning, and losses are concentrated in a specific, identifiable set of products. Together, these findings give management a clear, evidence-based basis for targeted action rather than broad, category-wide changes.